



EDUCATION

Master of Science, Robotics | Northeastern University, Boston, MA | GPA – 3.8/4 Dec 2024

Coursework: Autonomous Field Robotics, Mobile Robotics, Computer Vision, Control Systems, Robotics Sensing & Navigation.

Bachelor of Engineering, Electronics & Telecommunication | University of Mumbai, India | GPA – 8.4/10 Jun 2022

Coursework: Neural Networks & Fuzzy Logic, Image Processing & Machine Vision, Discrete Time Signal Processing.

EXPERIENCE

Robotics Research Assistant [incl. Co-op] | Field Robotics Lab, Northeastern University, Boston, MA Sep 2023 – Present

- Led the design and development of a multi-UAV swarm system for mapping dynamic scenes, with a dual RGB camera setup for stereo on each drone and ensuring precise time synchronization across all cameras and platforms.
- Developed a perception-based control algorithm with real-time camera feedback using *Python* and *ROS2* to achieve formation flight with collision avoidance for tracking dynamic targets using *Pixhawk 6X* and *ArduPilot*.
- Calibrated camera intrinsic and extrinsic parameters for the stereo-camera rig before each mission and used online calibration to get accurate depth estimates in real-time for optimal path planning, 3D reconstruction and mapping.
- Generated 3D/4D representations and novel views from multi-view datasets of dynamic scenes using *NeRFs* and *Gaussian Splatting*, from optimized sparse point clouds produced by Structure from Motion (SfM) algorithms.
- Benchmarked state-of-the-art Visual SLAM algorithms such as *NVIDIA ISAAC ROS VSLAM* on the on-board *Nvidia Jetson Orin Nano* to evaluate real-time performance on edge devices.

Graduate Teaching Assistant – Robotics Sensing and Navigation | Northeastern University, Boston, MA Sep 2023 – Dec 2024

- Guided students on data acquisition by writing device drivers in *ROS/ROS2*, post-processing and analysis techniques for IMU, GNSS, RTK GPS, Camera and LiDAR sensors.
- Led discussions on core robotics concepts such as localization, mapping, path planning, computer vision, optimization, etc. and their state-of-the-art algorithms like *ORB-SLAM*, *RTAB Map*, *LeGO-LOAM*, etc.

Computer Vision & IoT Intern | EdVerb Pvt. Ltd, Mumbai, India Jan 2022 – Jun 2022

- Programmed the *Arduino Uno* and *ESP-32* board and interfaced with multiple sensors to devise several cost-efficient computer vision and IoT-based solutions for automation.
- Designed and developed an automated parking management system using the *ESP-32 Cam* to implement automatic number plate detection and Optical Character Recognition (OCR) to register license plates.
- Deployed ultrasonic sensors for marking and determining slot availability at each parking space and continuously update the real-time database.

TECHNICAL SKILLS

Operating Systems & Software: ROS/ROS2 (Robot Operating System), Linux, Git, Docker, Gazebo, Raspberry Pi, Fusion 360.

Programming Languages & Libraries: Python, C++, MATLAB, OpenCV, PyTorch, CUDA, GTSAM, NumPy, Matplotlib.

Proficiencies: Robotics, Perception, SLAM, Deep Learning, System Design, Motion Planning, Machine Learning, CAD, 3D Printing.

PROJECTS

Semantic Segmentation of Road Cracks using Swin Transformer ([link](#))

- Trained a surface crack segmentation model for autonomous road inspection using Swin Transformer + UPerNet architecture, utilizing *PyTorch* and *MMSegmentation*, with ADE20K pre-trained weights for improved transfer learning.
- Achieved ~78% mIoU on the test set at 80k iterations by employing a hybrid loss strategy combining Cross-Entropy loss and Lovasz-Softmax loss to optimize IoU on fine-grained crack patterns across a dataset of 11.2K images.

3D Point Cloud Reconstruction from 2D Images through Structure from Motion (SfM) ([link](#))

- Developed a 3D reconstruction algorithm that performs Structure from Motion to generate a sparse point cloud from 2D images captured with a calibrated camera, using *OpenCV* for processing and *GTSAM* for optimization.
- Matched features between images using *SIFT* and *RANSAC* to estimate camera poses using essential matrix and Perspective-n-Point (PnP) pose computation, and triangulate their inlier feature points into 3D space.
- Optimized the point cloud by performing global Bundle Adjustment using *GTSAM*'s factor graphs with LM algorithm to improve camera poses and 3D landmarks from the initial triangulation, achieving a 94.5% decrease in reprojection error.

GPS & IMU Sensor Fusion using Extended Kalman Filter (EKF)

- Built and deployed custom *ROS2* based device drivers for real-time sensor data acquisition from a GNSS receiver and IMU for localization and navigation of a ground vehicle.
- Analyzed IMU noise using Allan Variance, calibrated magnetometer for hard/soft iron distortions, corrected accelerometer bias for velocity estimation, and refined heading by fusing gyroscope & magnetometer data for dead reckoning.
- Performed GPS and IMU sensor fusion by implementing Extended Kalman Filtering (EKF) to get an improved estimate of vehicle's overall trajectory including GPS-denied environments.

PUBLICATIONS

- A System for Multi-View Mapping of Dynamic Scenes Using Time-Synchronized UAVs, submitted to *IROS 2025*.